MILITARY SPECIFICATION

CONTROL-PANEL ASSEMBLY FOR PORTABLE HELIPORT ELECTRIC LIGHTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a weatherproof control-panel assembly for supplying and controlling the voltage for portable heliport lighting installations.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

DRAWINGS

ME

TA13217E7550	- Control Panel Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The control-panel assembly shall be as shown on Top Assembly TA13217E7550 and as specified herein. The control-panel assembly shall have one input connector receptacle for accepting 115/120 volts, ac, 60 cycles; one output connector plug; and three output connector receptacles. The output voltage at the output connector receptacles shall be individually controlled by autotransformers. The control-panel assembly shall be provided with a pilot light and a circuit breaker for protection of the autotransformers from overload. The components shall be enclosed with a splashproof, spraytight, metal case.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one control-panel assembly for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce control-panel assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Operation. Each control-panel assembly, with an input of 115 volts, ac, 60 cycles, and with the autotransformers set at maximum voltage, shall operate without malfunction when a load of two 100-watt, 110/115-volt lamps are connected parallel to any one of the four output circuits with no load on the other three circuits. The autotransformer's output voltage shall be not more than 120 volts and not less than 112 volts.

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3.5 Temperature and humidity. The control-panel assembly shall not be damaged by a storage temperature of 155°F with 80- to 98-percent relative humidity, and by operation at ambient temperatures from plus 120°F to minus 50°F with humidity uncontrolled.

3.6 Circuit breaker. The circuit breaker shall withstand a 15 percent overload for 60 minutes without overheating, and shall be set to trip at 25 percent overload within 60 minutes.

3.7 Splashproof and spraytight case. The case shall be splashproof and spraytight in accordance with MIL-STD-108.

3.8 Operational reliability. The control-panel assembly, connected to a 115-volt, ac, 60-cycle power source, and with a 5-ampere load, shall operate without malfunction for 50 hours.

3.9 Treatment and painting. The control-panel assembly shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, lusterless, color as specified (see 6.2).

3.10 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.10.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.10.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

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3.11 Identification marking. The identification markings shall be in accordance with MIL-STD-130, and as shown on the drawings.

3.12 Workmanship. The control-panel assembly shall be free from dirt, scale, flux, and other extraneous material. All electrical connections shall be secure, metal joints and seams shall be tight, and all edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction control-panel assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.1.

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4.3.2 Tests. The preproduction control-panel assembly shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.1.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level S-4.

4.4.2 Examination. Each control-panel assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each control-panel shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.4. AQL shall be 6.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The control-panel assemblies shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified.
102. Parts or components missing.
103. Parts or components misalined.
104. Materials or components not as specified.
105. Wiring and connections not as specified.
106. Treatment and painting not as specified.

Minor

201. Identification markings not as specified.
202. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Operation. Operate the control-panel assembly from a 115-volt, ac, 60-cycle power source. Set the autotransformer control dials on 100 volts. Connect two 100-watt, 110/115-volt lamps parallel to any one of the

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three circuits controlled by autotransformers. Actuate the master switch to the "ON" position. Set the appropriate autotransformer control dial to the maximum voltage position and operate for 10 minutes. Measure the output voltage. Reset the control dial to 100 volts. Repeat this test by alternately applying the lamp load to the other three output circuits. An output voltage of more than 120 volts, or less than 112 volts, or malfunction of the switch, shall constitute failure of this test.

4.5.2.2 Temperature and humidity. The chambers used for temperature and humidity shall maintain ambient temperatures within 4°F throughout the test.

4.5.2.2.1 Low-temperature storage and operation. Cold-soak the control-panel assembly at minus 50°F for a period of 12 hours. Test the control-panel assembly at maximum voltage for a period of 1 hour with one 100-watt lamp in each of the four circuits. At the end of each 15-minute period, adjust the autotransformers from maximum to minimum position and open and close the circuit breaker. Inability of the control-panel assembly to operate, or malfunction of any component, shall constitute failure of this test.

4.5.2.2.2 High-temperature and humidity storage and operation. Place the control-panel assembly in a test chamber with 80- to 98-percent relative humidity at an ambient temperature of plus 155°F for 24 hours; reduce the temperature to plus 120°F for 24 hours prior to start of operational tests. Check the circuit breaker by adjusting the output load to 15 percent overload (5.75 amperes) for 1 hour. Increase the output load to 25 percent overload (6.25 amperes) and determine the time required to trip the circuit open. Nonconformance to 3.4, damage to the breaker or other components during overload, or nonconformance to 3.6 shall constitute failure of this test.

4.5.2.3 Splashproof and spraytight. Test the control-panel assembly case in accordance with MIL-STD-108. Close and clamp the cover prior to start of test. Indication of moisture inside the control-panel assembly case shall constitute failure of this test.

4.5.2.4 Operational reliability. Operate the control-panel assembly from a 115-volt, ac, 60-cycle power source. Test the control-panel assembly with a 100-watt load on each of the four circuits for 50 hours at maximum voltage. Turn the circuit-breaker switch off and on manually after each 1-hour period. Malfunction of any component shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

Major

107. Materials, methods, and containers not as specified for Level A or B. Each incorrect material, method, or container shall constitute one defect.
108. Technical publications not contained in the case.

109. Case not blocked and braced within the container as specified for Level A or B.
110. Boxes not close-fitting for Level A or B.
111. Gross weight or size exceeds the limitation of the box specification for Level B.
112. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Each control-panel assembly with technical publications contained in the case shall be preserved in accordance with MIL-P-116, Method IA-14, using the case as the inner container and a close-fitting box conforming to PPP-B-636, W5c, style optional, as the outer container. The case shall be blocked and braced within the outer container to prevent movement and damage by the use of corrugated fiberboard pads fabricated from the same material as the box.

5.1.2 Level B. Each control-panel assembly with technical publications contained in the case shall be packaged in a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, Grade 175, style optional. The case shall be blocked and braced within the box in a manner to prevent movement and damage by the use of corrugated fiberboard pads fabricated from the same material as the box.

5.1.3 Level C. Each control-panel assembly with technical publications contained in the case shall be packaged to afford protection against damage from the supply source to the initial destination.

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5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The control-panel assemblies, packaged as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. The boxes shall be closed and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. The control-panel assemblies, packaged as specified in 5.1, shall be packed as specified in 5.2.1 except the boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c or boxes conforming to PPP-B-640, Class 2, style optional, may be used if the gross weight or size limitation of the box is not exceeded. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. The control-panel assemblies, packaged as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in fiberboard containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The control-panel assembly is intended to be used to regulate and adjust the voltage to the various circuits in portable heliport lighting installations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) Color required (see 3.9).
(d) Level of preservation and packaging and level of packing required (see 5.1 and 5.2). Level B preservation, packaging, and packing is intended to provide economical but limited protection and should be specified only when it is determined the control-panel assemblies will be held in covered storage.

6.3 Preproduction model. Any changes or deviations of production control-panel assemblies from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish control-panel assemblies conforming to this specification.

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6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each control-panel assembly.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of control-panel assemblies.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 6110-A180

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-C-52009C(ME) Control-Panel Assembly for Portable Heliport Electric Lighting

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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